CodeVision — System Architecture

AI-Powered Codebase Intelligence Platform · End-to-End System Design

Next.js 16 | Claude Sonnet | Tree-sitter | Supabase | TypeScript

28+	4	44	6
API ENDPOINTS	AI PASSES	TECH DETECTED	VISUAL VIEWS

Standard end-to-end system architecture

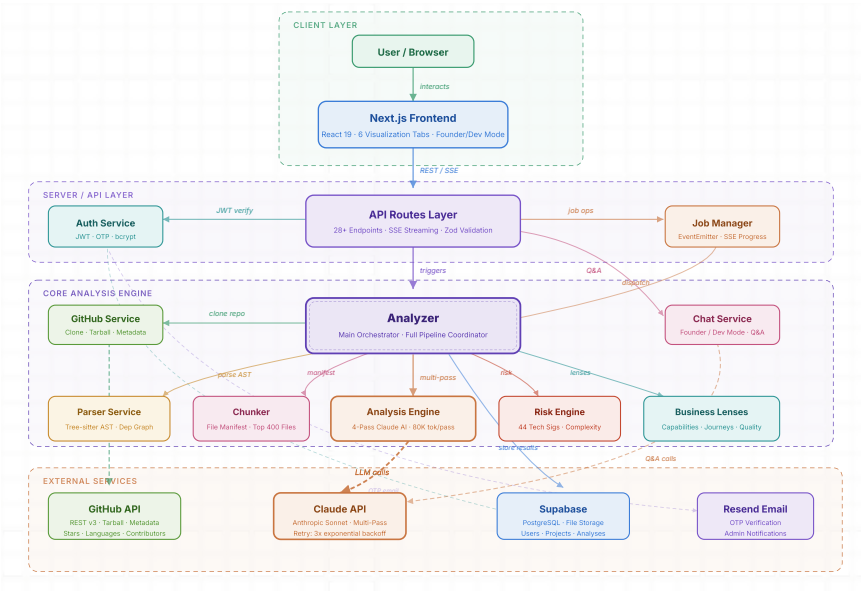

Figure 1. End-to-end system architecture of CodeVision. Each zone represents a logical tier; dashed edges indicate indirect or background communication. Click nodes in the interactive version at `/system-design`.

some more detail ..

Data processing pipeline — GitHub URL to structured intelligence

1 Repo Acquisition → 2 File Manifest → 3 AST Parsing → 4 Doc Parsing → 5 AI Analysis → 6 Risk Assessment → 7 Business Lenses → 8 Storage & Views

Tarball download or git clone. Fetches stars, languages, contributors, commit history.

Score & categorize files (entry_point=5, config=4, route=3). Select top 400 non-test files.

Tree-sitter for Python, JS, TS, Go, Java. Regex fallback. Builds dep graph + detects patterns.

PDF, markdown, image requirements downloaded from Supabase storage and extracted.

4-pass Claude pipeline. 80K tokens/pass. Retry with exponential backoff (max 3x).

44 tech signatures + static checks + AI-mapped risks. Complexity score 1-10. Cost: $900/day.

Capability graph (8 caps · 3 domains), journey graph, quality report with confidence scores.

6 UI-ready views compiled and stored. Project status set to *completed*.

The AI multi-pass analysis engine: how CodeVision progressively builds understanding

1 — PASS 1 Module Summaries

Analyzes the top 15 modules ranked by dependency degree — the most connected parts of the codebase. For each, it produces a plain-language business function, primary technologies, and estimated lines of code.

INPUT	Top modules + source files
OUTPUT	Business function per module
MODEL	Claude Sonnet · ≤80K tokens

2 — PASS 2 Architecture Graph

Generates 5–12 business-friendly nodes with typed edges (*imports, calls, stores, renders*) and 3–6 plain-English data flow steps. Uses module summaries and the dependency graph as context.

INPUT	Module summaries + dep graph
OUTPUT	Nodes · edges · data flow steps
MODEL	Claude Sonnet · ≤80K tokens

3 — PASS 3 Business Context

Bridges technical architecture and business outcomes. Extracts user journeys (4–6 steps each), value-driving features, data usage patterns, and gap findings vs. requirement documents.

INPUT	Architecture + requirements docs
OUTPUT	Journeys · features · gaps
MODEL	Claude Sonnet · ≤80K tokens

4 — PASS 4 Founder Narrative

Translates all prior outputs into a compelling, jargon-free narrative for non-technical stakeholders. Uses analogies, generates strategic insights, and competitive positioning summaries.

INPUT	All prior passes + full context
OUTPUT	Founder & technical narratives
MODEL	Claude Sonnet · ≤80K tokens

Client Layer · Server / API Layer · Core Analysis Engine · External Services

Data flow · External API call (async) · Indirect / background